

On the propagation factor $F_\star$: why it is not the quantity to report

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Abstract

This report examines the propagation factor $F_\star$ introduced in the red passages of `F_star/main_F_star.tex` and asks three questions: whether it can serve as the error bound the experiments report, what its evaluation costs relative to the reported factor $\hat{F}_\Sigma$, and whether it resolves the difficulty that a singular molecular potential creates for the supremum factor $F_{\Sigma+}$. The theoretical content of $F_\star$ is sound and its central observation is correct: the exponential dependence on the stage count K in the published constant is an artifact of the recursion, and telescoping over the injected errors removes it. What follows is not a criticism of that argument. The conclusion is that $F_\star$ is a good remark and a poor statistic. It is governed by a two-sided norm, and the side it adds — an essential infimum — is precisely the side a singular potential destroys, so it does not settle the singularity difficulty but inherits it twice. Its sample counterpart is an exponentiated extremum, which has no limit in the particle count, is a property of the realized genealogy rather than of the stage schedule, and is biased low at exactly the stages that dominate it. Finally, it cannot be computed from what the completed runs saved. Section 8 records the decision taken on the strength of this analysis: $F_\star$ is not mentioned anywhere in the paper.

1 What $F_\star$ is

Write $\mu_t \propto \exp(-U_t)$ with $U_t = (1-t)U_0 + tU$ for the interpolation from the source to the target, $\mu_k := \mu_{t_k}$ for the accepted stage targets, ν_k for the stage proposal, and $w_k := d\mu_k/d\nu_k$ for the stage weight. After the change of variables that removes the stage maps, the accumulated ratio

$$R_k := \prod_{j=k+1}^K r_j = \frac{d\varrho_K}{d\varrho_k}, \quad R_K := 1, \quad (1)$$

carries the stage- k target all the way to the endpoint in one reweighting. The proposed theorem assumes that each R_k is bounded on both sides off null sets of the source,

$$0 < \underline{R}_k := \text{ess inf } R_k \leq \text{ess sup } R_k =: \overline{R}_k < \infty, \quad \varrho_k^\star := \frac{\overline{R}_k}{\underline{R}_k} \geq 1, \quad (2)$$

and concludes the same bound as the published theorem with the factor

$$F_\star := \sum_{k=0}^K \varrho_k^\star \quad \text{in place of} \quad F_{\Sigma+} = \sum_{k=0}^K \prod_{j=k+1}^K 3\|w_j\|_\infty. \quad (3)$$

Two things are genuinely gained, and they should be stated before the objections. First, the geometric factor 3^{K-k} disappears. In the neutral case $\mu_k \equiv \mu$ with identity stage maps, every $R_k \equiv 1$ and $F_\star = K+1$, whereas $F_{\Sigma+} = (3^{K+1} - 1)/2$ and the constant grows like 9^K although

nothing whatever is lost by a stage. Second, the supremum of a product is replaced by the spread of a single accumulated ratio, and successive stage weights need not peak at the same configurations, so the replacement is typically far from an equality. The accompanying ordering lemma also shows that $F_\star$ dominates the accumulated factor F_{acc} and bounds the asymptotic variance *linearly* rather than through its square. All of this is correct and worth recording.

The rest of this report concerns whether the resulting quantity can be *reported*.

2 The hypothesis is two-sided, and the new side is the fragile one

The published theorem assumes only $\|w_k\|_\infty < \infty$. The proposed theorem assumes (2), which is strictly stronger in kind, not merely in degree: it adds a lower bound on a density. The red text acknowledges this — “the bound is vacuous when a stage weight comes arbitrarily close to zero on the carried source support” — and states plainly that “the two results are therefore not ordered in general”.

That non-ordering is decisive for a reported quantity. A factor whose finiteness is not implied by the finiteness of the factor it replaces cannot be substituted for it in the experimental tables: there are schedules on which $F_{\Sigma+} < \infty$ and $F_\star = \infty$. A reported column must be defined for every run in it.

It is worth being explicit about where the lower bound fails, because for the molecular targets it fails for a structural reason rather than an accidental one. Taking identity stage maps for clarity, the accumulated ratio of (1) is

$$R_k \propto \frac{\mu_K}{\mu_k} = \exp(-(1 - t_k)(U - U_0)), \quad (4)$$

so the two ends of R_k are governed by the two ends of the energy difference $U - U_0$:

- $\underline{R}_k = 0$ wherever $U - U_0 \rightarrow +\infty$, that is, in the repulsive core where the target potential diverges;
- $\overline{R}_k = \infty$ wherever $U - U_0 \rightarrow -\infty$, that is, in the far field where the Gaussian source potential diverges while the target potential does not.

The first item is exactly the singular-potential difficulty. The second is present for every stage $k < K$ whenever the source is Gaussian on an unbounded component and the target energy is capped. Both ends therefore fail for the same schedules on which $\|w_k\|_\infty$ was already in doubt, and the lower end fails for a reason the upper end did not have.

3 Does $F_\star$ settle the singularity error?

It does not. The relevant comparison is between norms. The manuscript’s Appendix B.3 already explains why F_Σ survives a singular potential and $F_{\Sigma+}$ does not: an effective sample size is an L^2 quantity and a supremum is an L^∞ quantity, and a square-integrable weight can have $\text{ESS}(w_k) > 0$ while $\|w_k\|_\infty = \infty$. The ordering of the three candidates by fragility is therefore

$$\underbrace{L^2}_{F_\Sigma} \prec \underbrace{L^\infty}_{F_{\Sigma+}} \prec \underbrace{L^\infty \text{ and } L^{-\infty}}_{F_\star},$$

and $F_\star$ sits at the fragile end. It asks for more control of the weight than the factor that already fails, so it cannot repair a failure caused by insufficient control.

The point survives even when both ends are finite. Suppose the domain is compact — a torus component, say — or that the regularized potential $U^{\{\rho\}}$ caps the energy at the threshold e , so that (2) holds. Then (4) gives

$$\varrho_k^* = \exp\left((1 - t_k)[\text{ess sup}(U - U_0) - \text{ess inf}(U - U_0)]\right) = \exp((1 - t_k) \text{osc}(U - U_0)). \quad (5)$$

The summand is the exponential of an accessible energy range measured in units of $k_B T$. For the molecular targets that range is not a small number: the regularization threshold e alone is a large multiple of $k_B T$, and the source and target potentials disagree by that order over the sampled region. A factor of the form $\exp(\text{hundreds})$ is not a diagnostic that a reader can compare between two runs; it is a number whose logarithm is the only readable part.

So the regularization, not F_* , is what makes the constant finite, and what it leaves is finite in the same sense that $\exp(300)$ is finite. This is the precise sense in which F_* does not settle the singularity error: it converts an infinite factor into an astronomically large one, by borrowing the cap that the sampling potential already applies, and it does so only on the end where the cap acts.

4 The sample counterpart is an exponentiated extremum

Even granting the hypothesis, the reported quantity would not be F_* but its sample counterpart. With S_k^i the accumulated suffix log-weight along the lineage of particle i , the proposal is

$$\hat{F}_* = \sum_{k=0}^K \exp\left(\max_i S_k^i - \min_i S_k^i\right). \quad (6)$$

Four properties of (6) disqualify it as a reported number.

It has no limit in N . A maximum and a minimum over N draws are extremum statistics. For a log-weight with unbounded support the observed range $\max_i S_k^i - \min_i S_k^i$ grows without bound as N increases, so $\hat{F}_*$ increases with the particle count and does not converge. It therefore measures the sample size as much as it measures the run. By contrast $\text{ESS}(\mathbf{W}_j)$ is a second moment: it converges as N grows, and $\hat{F}_\Sigma$ inherits that stability. The manuscript reports factors computed at N between 2.4×10^5 and 3.2×10^5 and compares them across molecules; a statistic that drifts with N cannot be used that way.

The drift is then exponentiated. The quantity that grows with N sits inside an exponential. An increase of the observed log-range by 5, which a tenfold increase in N can easily produce for a heavy-tailed weight, multiplies the reported factor by about 150. The reported value is thus not merely unstable but unstable on a multiplicative scale that swamps any difference between two training objectives, which is the comparison the factor exists to make.

It is a property of the genealogy, not of the schedule. The accumulated ratio $\hat{R}_k^i = \prod_{j>k} W_j^{a_j(i)}$ is taken along the ancestral line $a_j(i)$ produced by the realized resampling draws. Two inference passes with the same trained flows, the same accepted schedule and different resampling seeds produce different lineages and therefore different $\hat{F}_*$. The published factors do not have this property: $\hat{F}_\Sigma$ and $\hat{F}_{\Sigma+}$ are functions of the stage weight vectors alone. A factor that characterizes “one accepted schedule”, which is how the manuscript uses the term, must not depend on the resampling draws of a particular replay.

Its bias runs the wrong way. The red text records the defect itself: resampling makes lineages coalesce, so the number of distinct ancestors at stage k falls as $K - k$ grows, and the observed spread of $\hat{R}_k^i$ for early k is computed over few distinct values and is biased low. The early stages are precisely the ones with the largest accumulated ratio and therefore the largest summands in (6). The estimator is thus optimistic exactly where it matters most, and its bias cannot be signed away by increasing N , since coalescence is driven by the number of stages.

5 The calculation is more expensive, and the saved runs cannot support it

The cost argument is concrete rather than asymptotic.

What the reported factors need. $\hat{F}_\Sigma$ and $\hat{F}_{\Sigma+}$ are read off the stage weight vectors $\mathbf{W}_j = (W_j^1, \dots, W_j^N)$, which every run evaluates anyway: one second moment and one maximum per vector, a single pass over data already on disk. The completed chiral runs persist exactly these, per stage: `flow_log_weights.npy`, `sharpen_log_weights.npy`, `samples.npy`, the two acceptance arrays, and `metadata.json`.

What $F_\star$ needs, and what is missing. Equation (6) needs the resampling ancestor indices $a_j(i)$. No ancestor, index or lineage array appears anywhere under the chiral run directories. The saved artifacts therefore do not determine $\hat{F}_\star$, and no amount of post-processing recovers it: the genealogy was not recorded and cannot be reconstructed from weights and positions. Reporting $F_\star$ for the completed runs would require re-running inference on all three molecules with ancestry recorded.

The coalescence-free alternative is a second training-scale replay. The remedy offered for the coalescence bias is to push each stage- k validation particle through the remaining maps $G_{k+1}^{-1}, \dots, G_K^{-1}$ without resampling. This is $O(K)$ flow evaluations per particle per starting stage and $O(K^2N)$ in total. For alanine dipeptide at $K = 10$ and $N = 3.0 \times 10^5$ that is on the order of 10^7 flow evaluations, against the $O(KN) \approx 3 \times 10^6$ of a single inference replay: a factor of K more work than the pass that produced the samples in the first place, to obtain one column of one table.

And the summary side is also unavailable. The per-stage summaries saved with the reported factors record the effective sample size and the largest rescaled weight, but not the smallest. The infimum side of (2), which is the side $F_\star$ adds, was never persisted for any of the three molecules.

6 What the measured runs show

Table 1 reproduces the factors computed from the three completed chiral runs. These are measured values from the saved artifacts, not estimates.

Two features of these runs bear on the question. First, the spread between the two published factors is already 18 to 29 orders of magnitude, and the whole of that spread is the product structure that $F_\star$ is designed to remove. The diagnosis in the red text is therefore correct on the real runs and not only on the six-state illustration: the exponential growth is an artifact. Second, the per-stage maxima that generate it are individually modest — the largest rescaled weight over all entries of all three runs is 106.07, at alanine dipeptide stage 2 — while the corresponding effective sample

molecule (d)	N	stages	$\hat{F}_\Sigma$	$\hat{F}_{\Sigma+}$	$\log_{10} \hat{F}_{\Sigma+}$
(2 <i>R</i> , 3 <i>R</i>)-2,3-butanediol (42)	2.4×10^5	8	37.45	1.01×10^{20}	20.01
L-alanine dipeptide (60)	3.0×10^5	10	68.08	3.55×10^{26}	26.55
Ac-Pro-NHMe (72)	3.2×10^5	11	176.75	3.37×10^{31}	31.53

Table 1: Propagation factors of the three completed chiral runs, accumulated over the selected flow or identity weight of every accepted stage followed by its sharpening weight. Reproduced from the saved run summaries.

sizes never fall below 0.410. A schedule whose worst stage retains 41% of its particles is not a pathological one, and yet its supremum factor is 10^{26} . This is the strongest available evidence that the exponential constant should be read as an artifact, and it is an argument for the remark, not for a new reported column.

7 Conclusion and recommendation

1. **The mathematics of $F_\star$ is correct and its diagnosis is right.** The telescoping identity, the theorem, and the ordering lemma all hold as stated, and the exponential dependence on K in the published constant is genuinely an artifact of re-entering a worst case at every stage.
2. **It is not a reportable error bound.** Its hypothesis is two-sided and is not implied by the hypothesis it replaces, so the two theorems are not ordered and $F_\star$ can be infinite where $F_{\Sigma+}$ is finite.
3. **It does not settle the singularity difficulty.** It requires control of the weight in L^∞ and of its reciprocal, so it is strictly more fragile than the factor that already fails on a singular potential. Where the regularized potential does make it finite, the value is $\exp(\text{osc}(U - U_0))$ by (5), an exponential of an energy range in units of $k_B T$.
4. **Its estimator is unsuitable independently of all that.** It is an exponentiated extremum: it drifts upward with N , it depends on the realized resampling genealogy rather than on the accepted schedule, and its coalescence bias is low at the dominant stages.
5. **It costs a second replay, and cannot be evaluated at all on the completed runs,** which persisted neither ancestor indices nor the minimum-weight side of the per-stage summaries.

The reported factor should therefore remain $\hat{F}_\Sigma$, which is bounded below by $K + 1$, built from second moments that converge in N , computable from vectors already on disk, and independent of the resampling draws. The manuscript’s existing position — that $\hat{F}_\Sigma$ is not an error bound but is the honest summary of accumulated stagewise weight degeneracy along one accepted schedule — is the one the evidence supports.

8 Decision

Decided by Xuda Ye, 2026-08-02: $F_\star$ is not to be mentioned at all, anywhere in this paper.

The decision is one of scope rather than of correctness. This is a computational frontier paper: it proposes a training objective and a generator, and it reports what a run costs and what it recovers. The propagation factor enters it as a summary of accumulated stagewise weight degeneracy, not as the subject of the paper. A second factor, a second theorem, its two-sided hypothesis, the ordering lemma relating the two, and the discussion of which is honest in which direction would add a strand of analysis that the paper’s readership is not reading it for, and that no experiment in it uses. The material in `main_F_star.tex` is therefore held outside the manuscript in its entirety, including the parts of it that are correct.

Three consequences are worth recording so that the decision is not partially undone later.

- **Nothing in the manuscript needs to change.** $F_\star$, $\varrho_k^\star$, F_{acc} , the accumulated ratio and the two-sided hypothesis appear nowhere in `Paper/main.tex`. The symbols R_k and essinf that do occur there are a different object and a different use: $R_k^i = r_k(Z_{k-1}^i)$ is the reduced-procedure stage weight of Appendix B, and essinf appears only in the definition $\text{osc}(\varphi) = \text{ess sup } \varphi - \text{ess inf } \varphi$ in Section 4.4.
- **The pessimism in K is already stated without $F_\star$.** Appendix B.3 makes that point on its own terms, with the neutral-case computation $F_{\Sigma+} = (3^{K+1} - 1)/2$ and a constant growing like 9^K , and concludes that the theorem establishes the rate $N^{-1/2}$ and the finiteness of the constant rather than its size. That paragraph needs no reference to a second factor, and none should be added to it.
- **No column, remark, footnote or appendix.** The exclusion covers the reported tables, the discussion of what the factor does and does not measure, and any forward-looking sentence promising a sharper constant. A partial mention would carry the cost the decision is avoiding — a reader asking what the other factor is — without the benefit of the full treatment.

The analysis in Sections 1–7 above is retained here, and only here, as the record of why the question was asked and how it was answered. Should a later and more theoretical paper take up the constant itself, this report and `main_F_star.tex` are the starting point: the telescoping argument, the theorem, and the ordering lemma are sound, and the objections in Sections 2–6 are objections to reporting $F_\star$ in *this* paper, not to the result.